

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0949

Calectric 138 kV
November 25, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0949. The Interconnection Customer proposes a 97.4 MW wind facility interconnecting at the Calectric 138 kV substation in the West Virginia area. The request was received on February 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 306 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$74,099,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Iron Grove – Waterton 138 KV
Contingency: Rockaway Beach – Mattoon East 138 KV
Loading: 198.9% (Rating: 177 MVA)
DFAX: 5.2%
Pre-project loading: 196.0%
2. Facility: Hampshire Paper – Waterton 138 KV L
Contingency: Rockaway Beach – Mattoon E 138 KV
Loading: 194.7% (Rating: 177 MVA)
DFAX: 5.2%
Pre-project loading: 191.8%
3. Facility: Panda Substation – Goat Rock 138 KV
Contingency: Panda – Northern Steel Casting 138 KV
Loading: 194.6% (Rating: 188 MVA)
DFAX: 10.8%
Pre-project loading: 189.0%
4. Facility: Northern Steel Casting – Goat Rock 138 KV
Contingency: Panda – Northern Steel Casting Substation 138 KV L
Loading: 186.6% (Rating: 199 MVA)
DFAX: 10.8%
Pre-project loading: 181.3%
5. Facility: Panda – Northern Teel Casting 138 KV
Contingency: Northern Steel Casting – Goat Rock 138 KV
Loading: 181.6% (Rating: 226 MVA)
DFAX: 11.9%
Pre-project loading: 176.5%

6. Facility: Carlsbad Waterfield – Menlo Substation 138 KV
Contingency: W Woodland – Menlo 138 KV line
Loading: 174.5% (Rating: 205 MVA)
DFAX: 10.7%
Pre-project loading: 169.4%
7. Facility: Carlsbad Waterfield – E. Ripley 138 KV Line
Contingency: W Woodland – Menlo 138 KV Line
Loading: 160.4% (Rating: 219 MVA)
DFAX: 10.7%
Pre-project loading: 155.7%
8. Facility: E. Ripley – West Fork 138 KV
Contingency: Menlo – West Woodland Substation 138 KV L
Loading: 151.0% (Rating: 227 MVA)
DFAX: 10.7%
Pre-project loading: 146.4%
9. Facility: Tank Hill – Eules 138 KV
Contingency: Carlsbad Waterfield – Menlo 138 KV
Loading: 120.2% (Rating: 230 MVA)
DFAX: 8.8%
Pre-project loading: 116.5%
10. Facility: Eules – W Woodland 138 KV
Contingency: Carlsbad Waterfield – Menlo 138 KV
Loading: 120.1% (Rating: 233 MVA)
DFAX: 8.8%
Pre-project loading: 116.4%
11. Facility: Panda – Menlo 138 KV line
Contingency: Fishkill Plains – Dillard-Orange 345 KV
Loading: 120.0% (Rating: 383 MVA)
DFAX: 12.6%
Pre-project loading: 116.8%
12. Facility: W Woodland Substation – Menlo 138 KV
Contingency: Carlsbad Waterfield – Menlo 138 KV
Loading: 119.5% (Rating: 237 MVA)
DFAX: 8.8%
Pre-project loading: 115.9%
13. Facility: Hampshire Paper – Rockaway Beach 138 KV line
Contingency: Waterton – Iron Grove 138 KV
Loading: 119.0% (Rating: 303 MVA)
DFAX: 15.9%
Pre-project loading: 113.9%
14. Facility: Northern Steel Casting – Solana Substation 138 KV
Contingency: Fishkill Plains – Dillard-Orange 345 KV
Loading: 118.9% (Rating: 397 MVA)
DFAX: 12.6%
Pre-project loading: 115.8%
15. Facility: Mattoon East – Rockaway Beach 138 KV
Contingency: Waterton – Iron Grove 138 KV
Loading: 118.6% (Rating: 307 MVA)
DFAX: 15.9%
Pre-project loading: 113.5%
16. Facility: Solana – Hampshire Paper 138 KV
Contingency: Fishkill Plains – Dillard-Orange 345 KV

Loading: 118.4% (Rating: 403 MVA)

DFAX: 12.6%

Pre-project loading: 115.4%

17. Facility: Iron Grove – Union Vly 138 KV Line

Contingency: Mattoon East Substation – Rockaway Beach 138 KV

Loading: 113.2% (Rating: 204 MVA)

DFAX: 9.6%

Pre-project loading: 108.7%

18. Facility: Union Valley – Potomac Yard 138 KV Line

Contingency: Mattoon E – Rockaway Beach Substation 138 KV

Loading: 108.4% (Rating: 251 MVA)

DFAX: 16.5%

Pre-project loading: 102.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waterton – Iron Grove 138 KV	Rebuild line with higher-capacity conductor	\$4,188,000	100% (\$4,188,000)
Waterton – Hampshire Paper 138 KV	Rebuild line with higher-capacity conductor	\$3,567,000	100% (\$3,567,000)
Panda Substation – Goat Rock 138 KV	Rebuild line with higher-capacity conductor	\$9,392,000	100% (\$9,392,000)
Northern Steel Casting Substation – Goat Rock 138 KV Line	Rebuild line with higher-capacity conductor	\$4,467,000	100% (\$4,467,000)
Panda – Northern Teel Casting 138 KV	Rebuild line with higher-capacity conductor	\$2,849,000	24% (\$672,000)
Carlsbad Waterfield – Menlo 138 KV	Rebuild line with higher-capacity conductor	\$7,995,000	100% (\$7,995,000)
Carlsbad Waterfield – E. Ripley 138 KV	Rebuild line with higher-capacity conductor	\$8,367,000	100% (\$8,367,000)
E. Ripley – West Fork 138 KV	Rebuild line with higher-capacity conductor	\$4,827,000	100% (\$4,827,000)
Tank Hill – Eules 138 KV	Rebuild line with higher-capacity conductor	\$4,586,000	100% (\$4,586,000)
Eules – W Woodland 138 KV	Rebuild line with higher-capacity conductor	\$3,904,000	100% (\$3,904,000)
Menlo – Panda 138 KV	Rebuild line with higher-capacity conductor	\$8,359,000	23% (\$1,900,000)
Menlo – West Woodland 138 KV	Rebuild line with higher-capacity conductor	\$6,314,000	100% (\$6,314,000)
Hampshire Paper – Rockaway Beach 138 KV line	Rebuild line with higher-capacity conductor	\$9,724,000	27% (\$2,606,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Solana – Northern Steel Casting 138 KV	Rebuild line with higher-capacity conductor	\$6,435,000	23% (\$1,462,000)
Mattoon East Substation – Rockaway Beach 138 KV	Rebuild line with higher-capacity conductor	\$11,666,000	27% (\$3,127,000)
Solana Substation – Hampshire Paper 138 KV	Rebuild line with higher-capacity conductor	\$5,409,000	23% (\$1,229,000)
Union Vly – Iron Grove 138 KV	Rebuild line with higher-capacity conductor	\$4,960,000	100% (\$4,960,000)
Union Valley – Potomac Yard 138 KV Line	Reconductor line	\$2,510,000	21% (\$536,000)
	Total allocated to project		\$74,099,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.